Assignment 9: Double Damascene Process

In this report, our analysis will centre on vias and trenches, which are essential for establishing electrical connections within the a device. We will also explore the materials employed in these structures, including intermetal dielectrics (IMDs), etch stop layers, and metal fills. By comparing various process flows and examining a 14nm logic process, we aim to gain a comprehensive understanding of structure.

Question A: Comparison of both processes

This part of assignment shines light on difference between trench first and via first process along with key challenges in both processes.

Both processes have its own advantage and disadvantage, trench first have benefit of creating shallower trenches whereas via first have simpler mask design and deeper vias.

Following figures shows process of trench first and via first process that was followed,

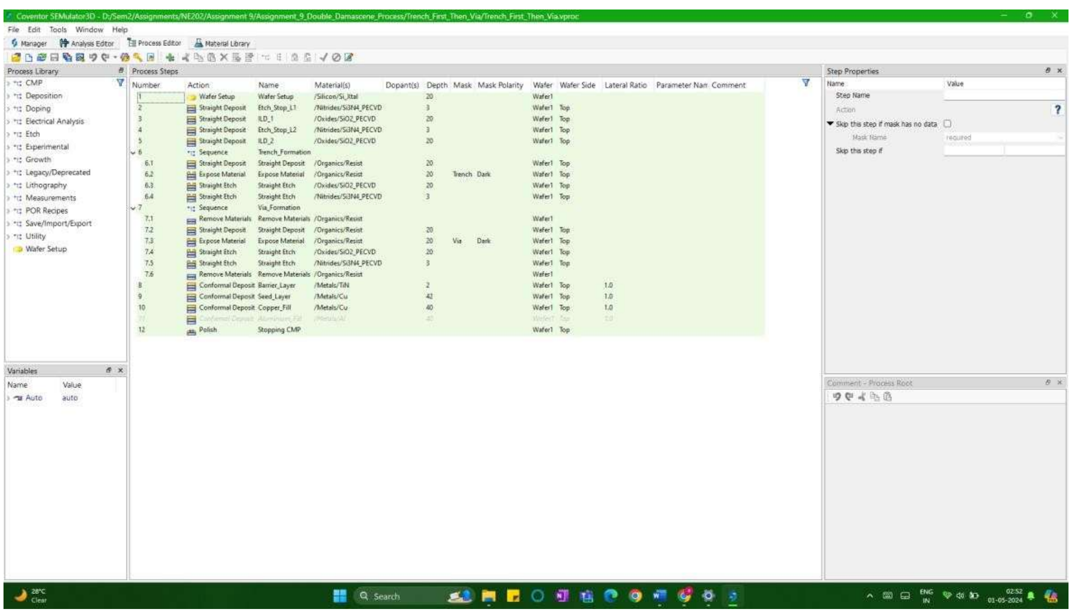


Figure 1 : Process flow of Trench first process

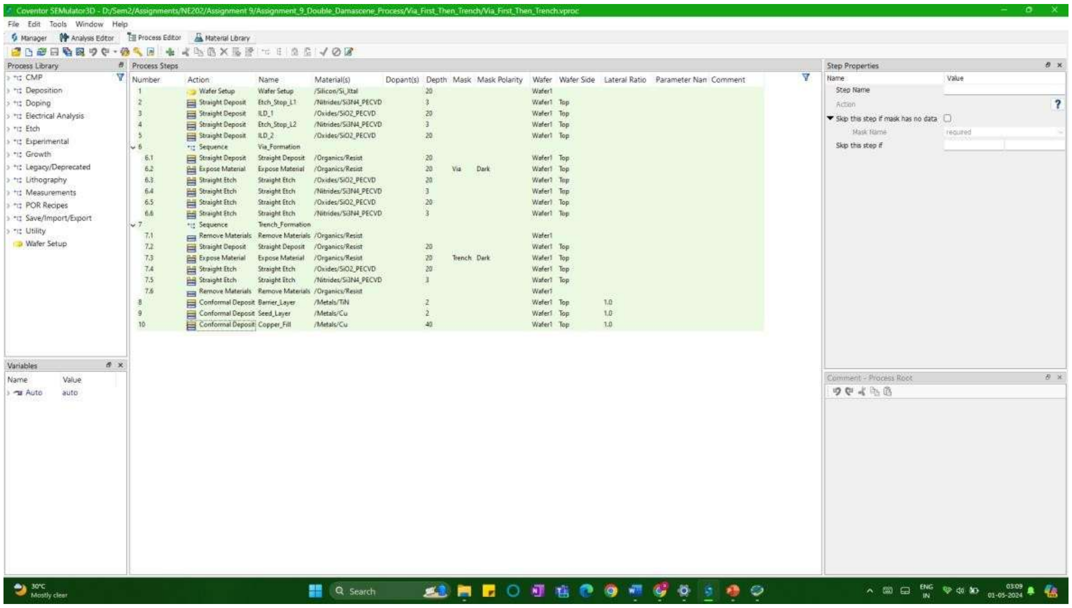


Figure 2 : Process flow of Via first process

Following figures shows simulated results of trench first and via first process that was followed,

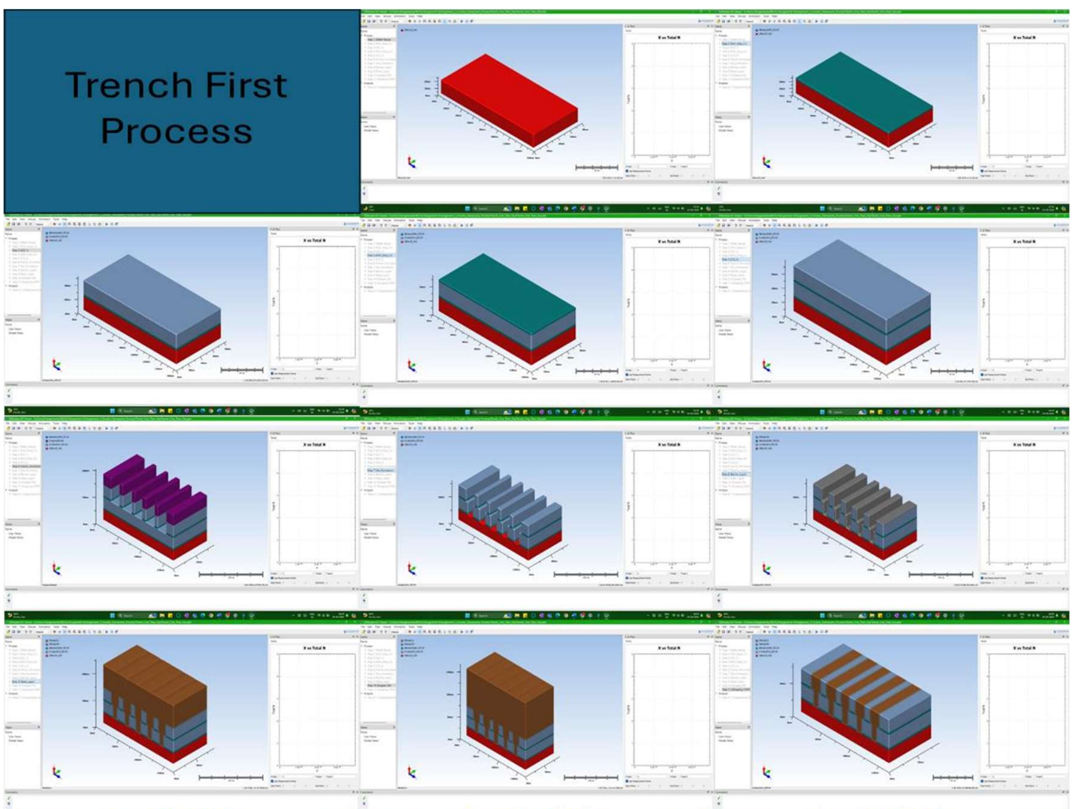


Figure 3 : Simulated results for Trench first process

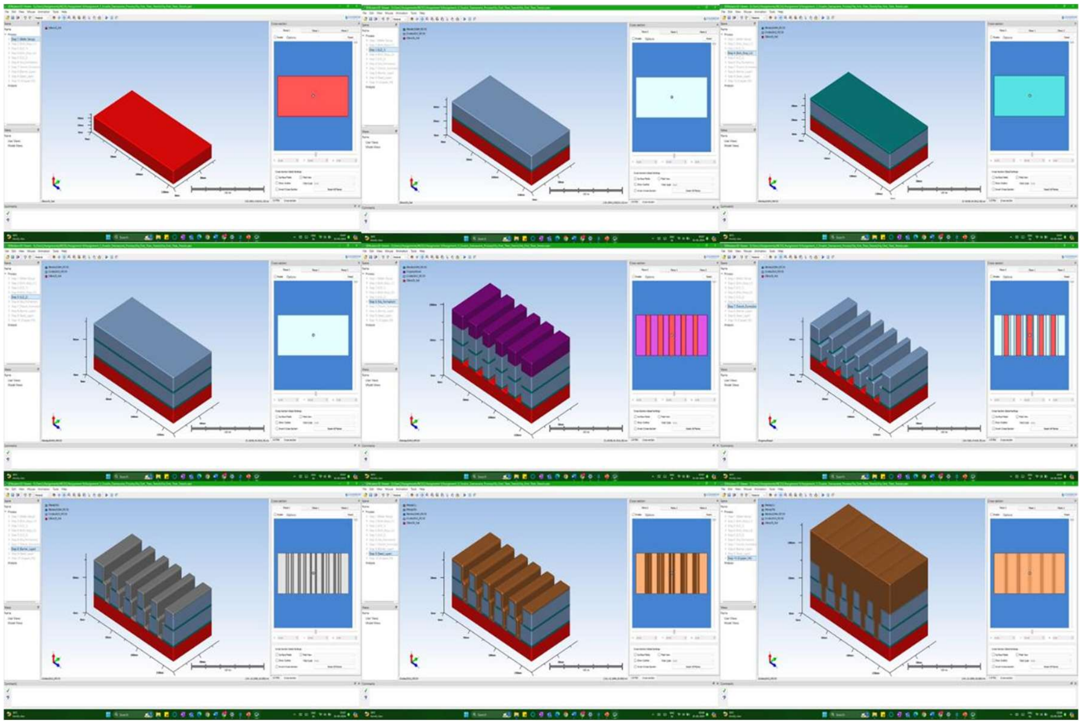


Figure 4 : Simulated results for Via first process

Question A(i)

Difference in trench first and via first process is as below:

1. **Trench First:** This approach needs a photoresist (PR) layer deposited inside the pre-etched trenches to define the via patterns. In contrast, the Via First process uses a simpler mask directly on a flat surface for via patterning.
2. **Via First:** This method creates vias with a larger depth-to-width ratio (aspect ratio) compared to Trench First. Via First involves etching deeper vias individually, whereas Trench First etches the insulating material (ILD) between trenches in separate steps.

Question A(ii)

The challenges in both process flows:

- It's tricky to define via patterns inside pre-made trenches because of the uneven surface. This makes it harder to create very precise vias using lithography. Via First, on the other hand, works on a flat surface, making via patterning easier and more accurate.

- Via first method creates deeper vias, which can be a problem as ICs get smaller. Etching these deeper vias sideways can limit how many vias you can fit close together.

Question B: Inter-Metal Dielectric (IMD)

Inter-Metal Dielectric is used to isolate two metal lines as it is a dielectric in between two metals it forms a capacitance.

We will analyse the material used for IMD and find capacitance, dielectric constant and potential profile due to this.

Question B(i)

The material used for **IMD is SiO_2_PECVD**.

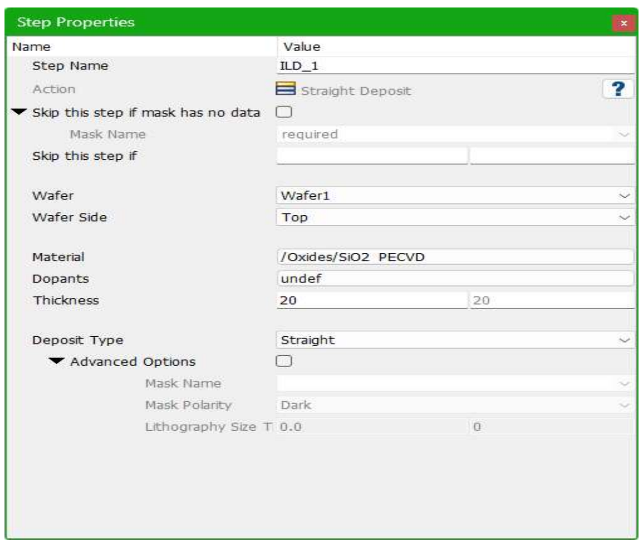


Figure 5 : IMD used in process

To measure capacitance value, we utilize the analysis window and calculate capacitance between (19,40,56) and (42,40,56).

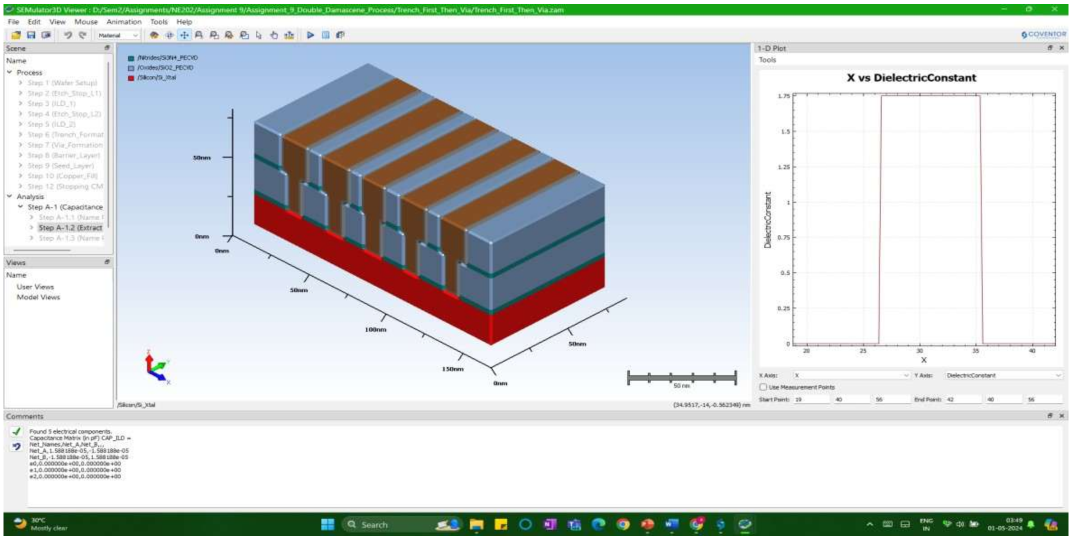


Figure 6 : Capacitance Measurement

Measured capacitance value is **C = 1.588188e-05 pF**.

Question B(ii)

The IMD material was changed from SiO_2 to Si_3N_4 as shown in figure below,

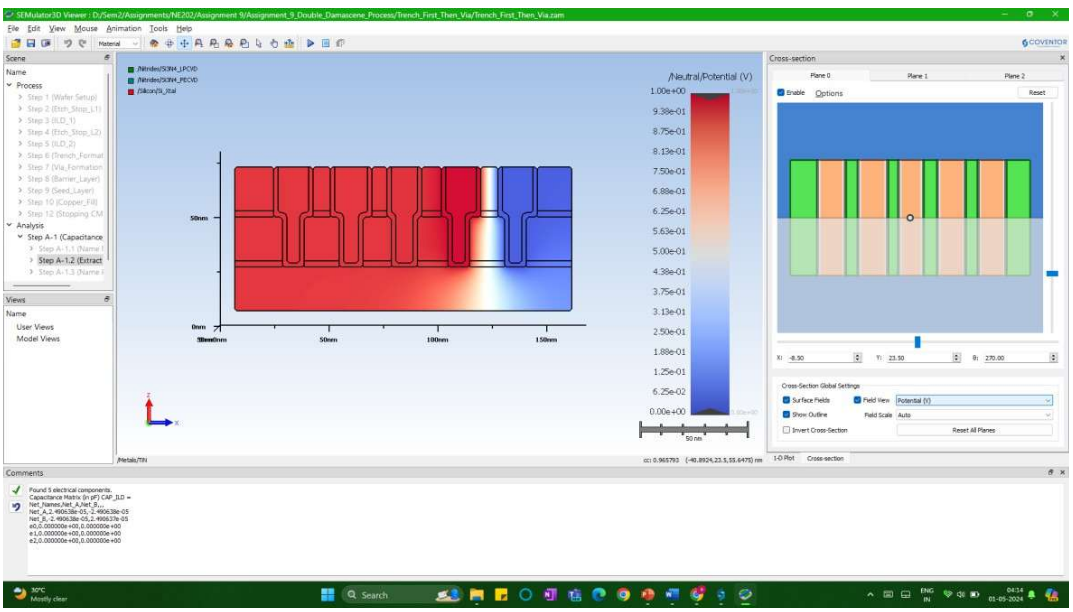


Figure 7 : Changing IMD material to Si_3N_4

Measured capacitance value is **C = 2.490638e-05 pF.**

Dielectric constant and potential profile are shown in the figure below,

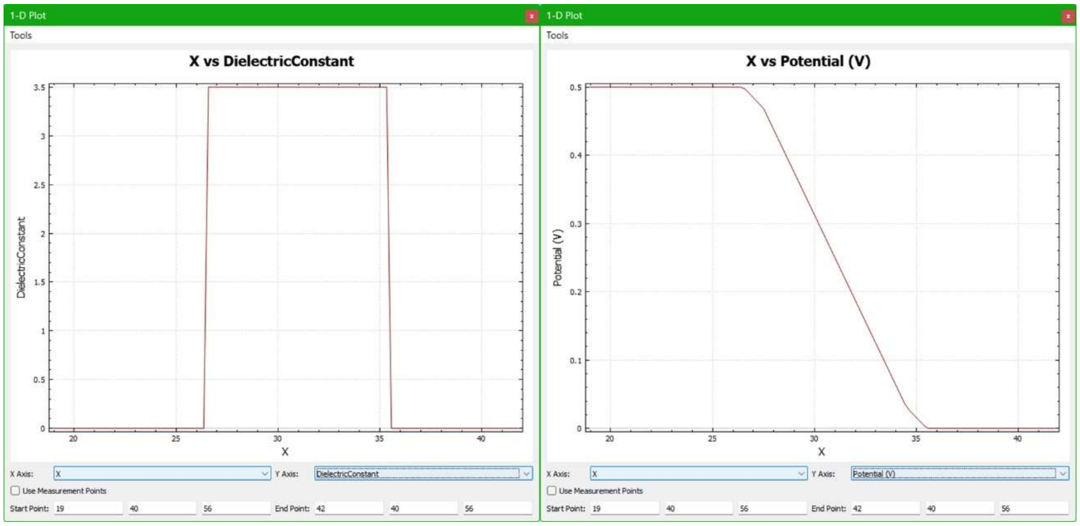


Figure 8 : Dielectric constant and Potential Profile for Si_3N_4

The IMD material was changed from SiO_2 to HfO_x as shown in figure below,

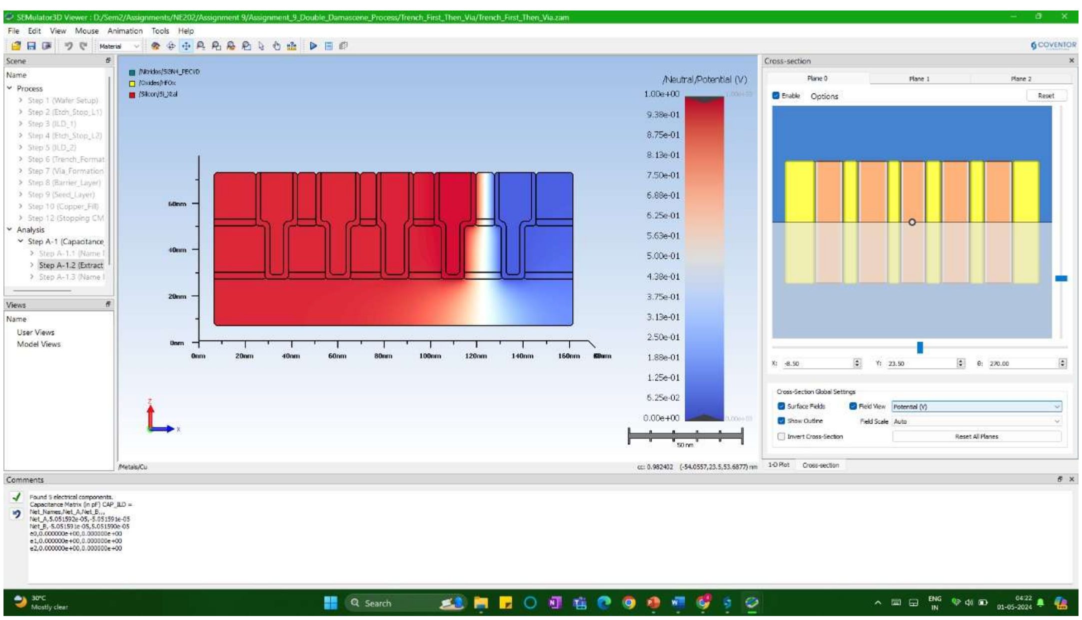


Figure 9 : Changing IMD material to HfO_x

Measured capacitance value is **C = 5.051592e-05 pF.**

Dielectric constant and potential profile are shown in the figure below,

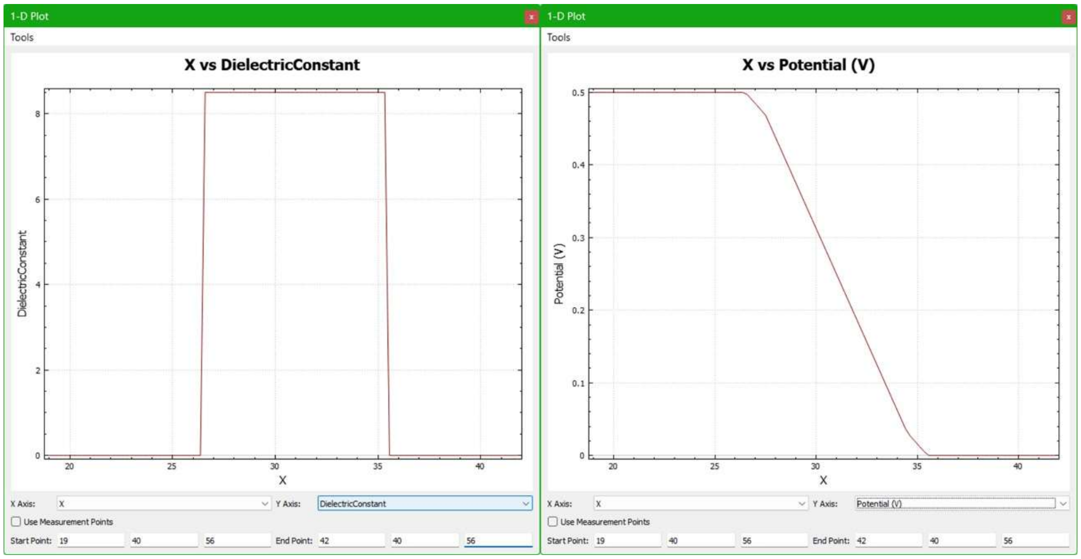


Figure 10 : Dielectric constant and potential profile for HfO_x

To ensure fast signal transmission and minimize the effects of capacitance, we prefer materials with a low k-value. However, there's a trade-off. The k-value can't be too low either. It needs to be high enough to prevent electrical breakdown between the metal lines.

Here, among all SiO_2 ($k < 1.75$) works best as IMD material.

Question C: Etch stop layer

Here, **Si_3N_4** is used as etch stop layer.

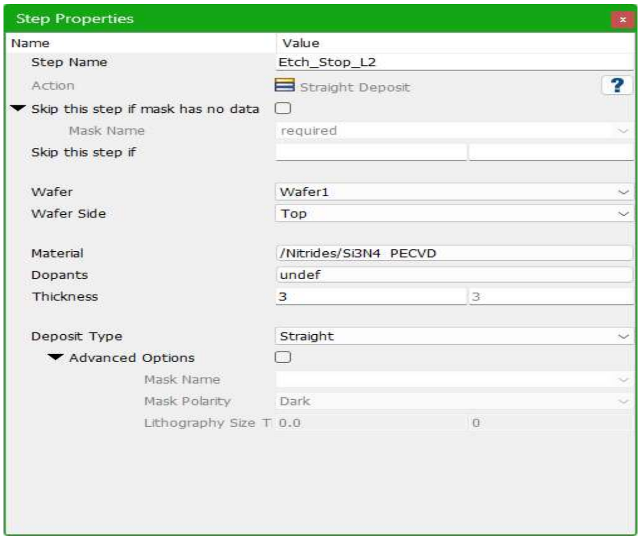


Figure 11 : Etch stop layer used in process

There are two etch profiles one for trench formation and another for via formation, for both the model uses Si_3N_4 as barrier layer to prevent etching beyond this layer. Figure below shows the etching profile.

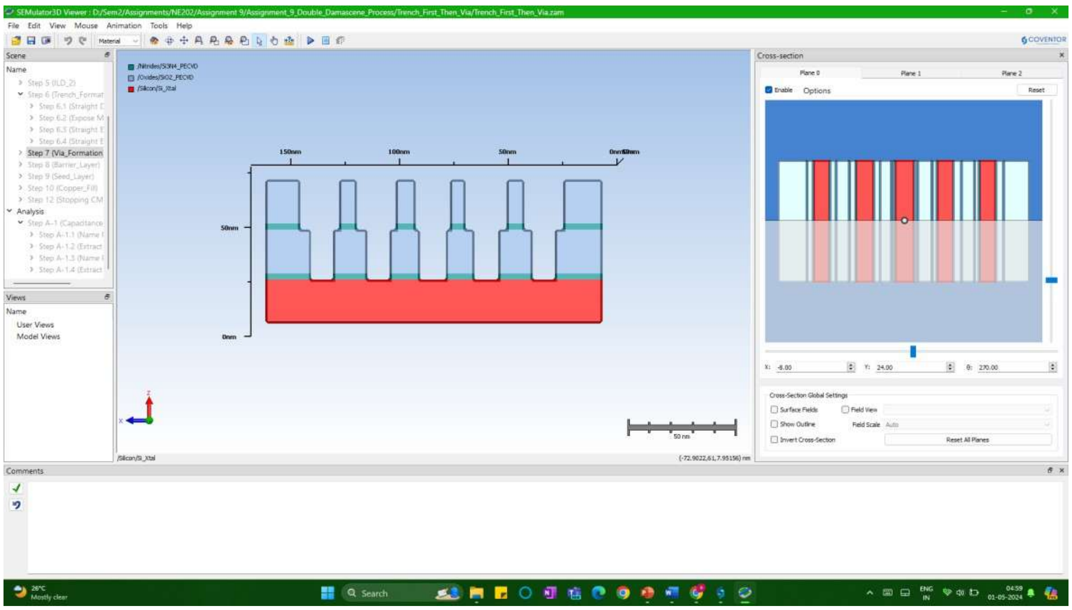


Figure 12 : Etch profile after via formation

Question D: Filler metal

The model has copper as default filler metal with resistance of **5.061142e-02 Ohms** and various electrical profiles as shown by figures below,

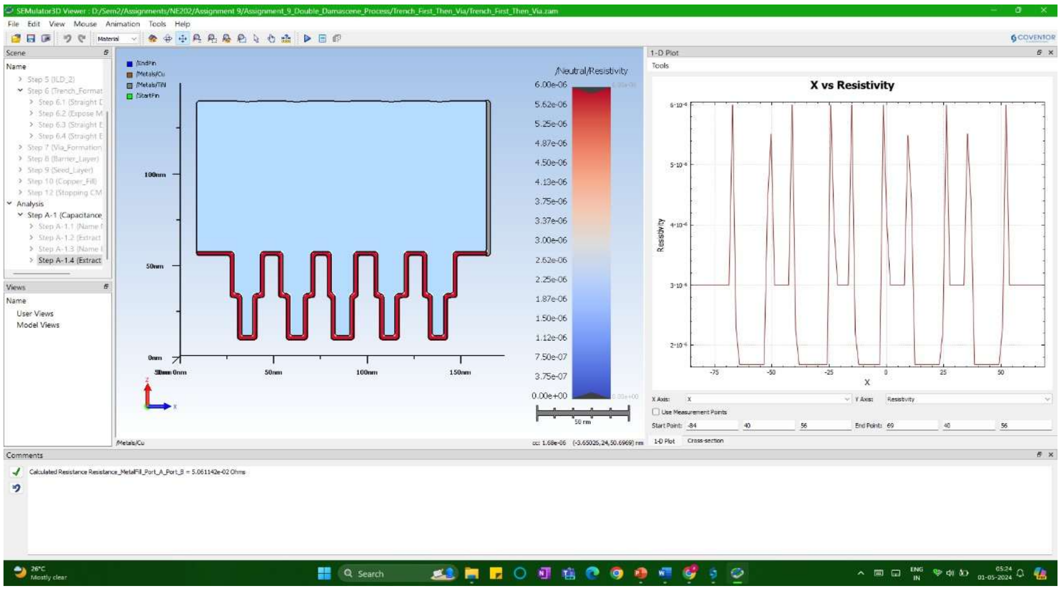


Figure 13 : Resistivity profile for Cu

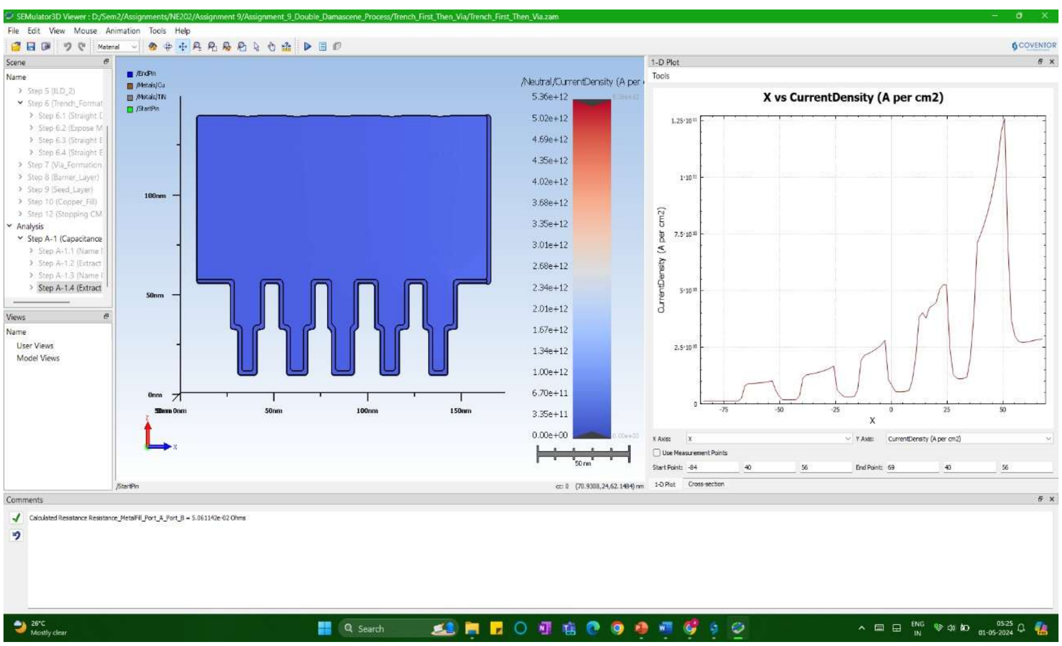


Figure 14 : Current density profile for Cu

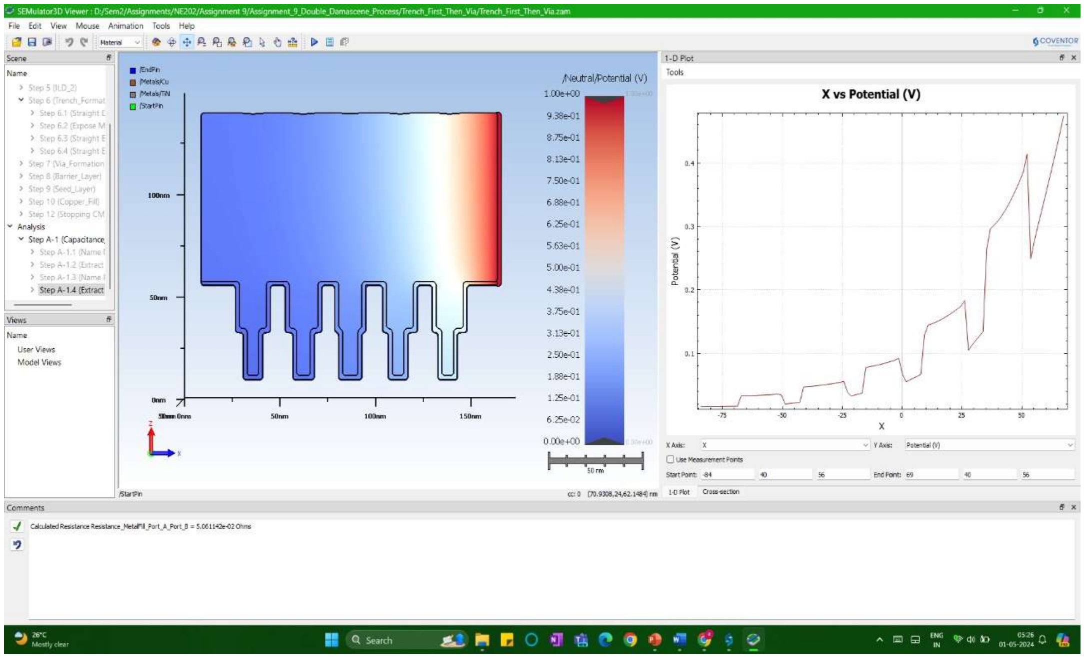


Figure 15: Potential profile for Cu

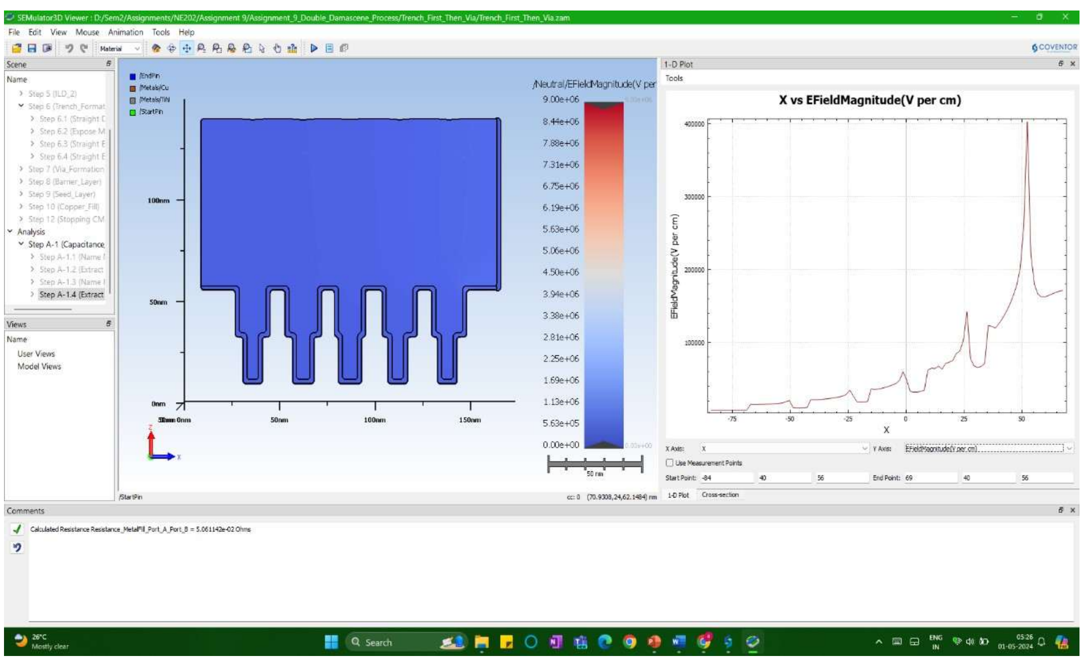


Figure 16: Electric field profile for Cu

Changing filler metal to aluminium gives resistance value of **7.945796e-02 Ohms** and profiles as shown in figures below,

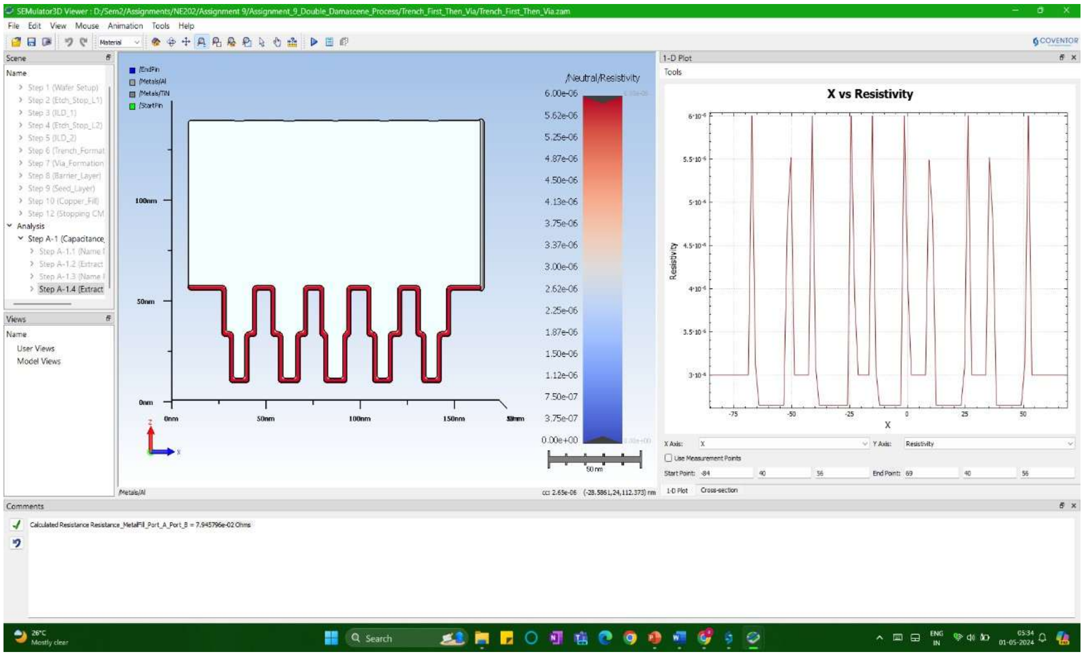


Figure 17: Resistivity profile for Al

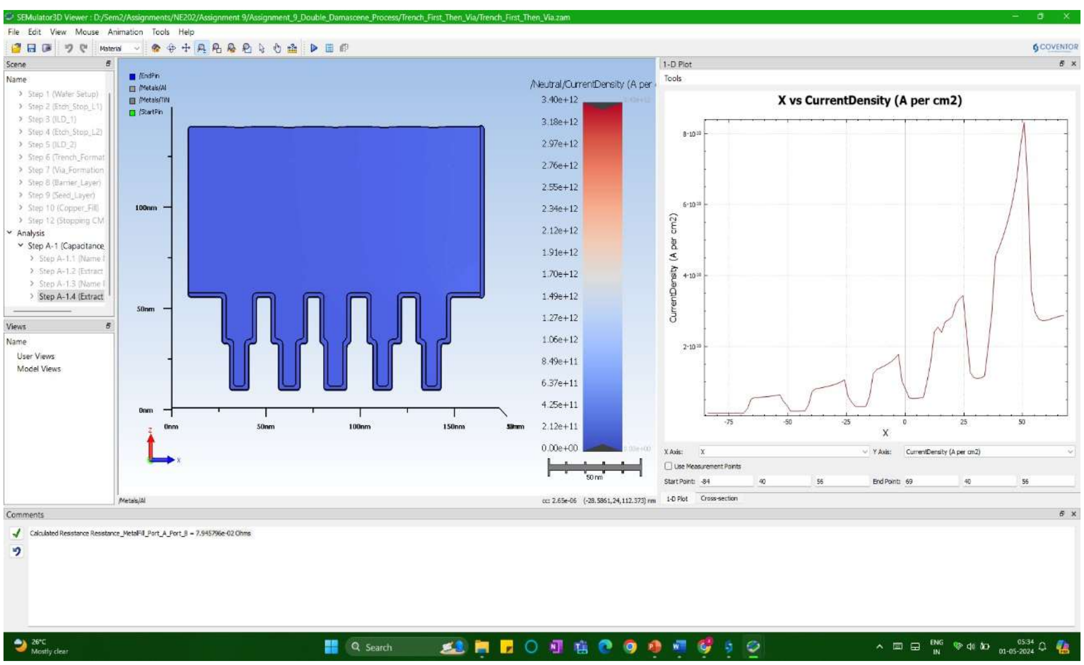


Figure 18: Current density profile for Al

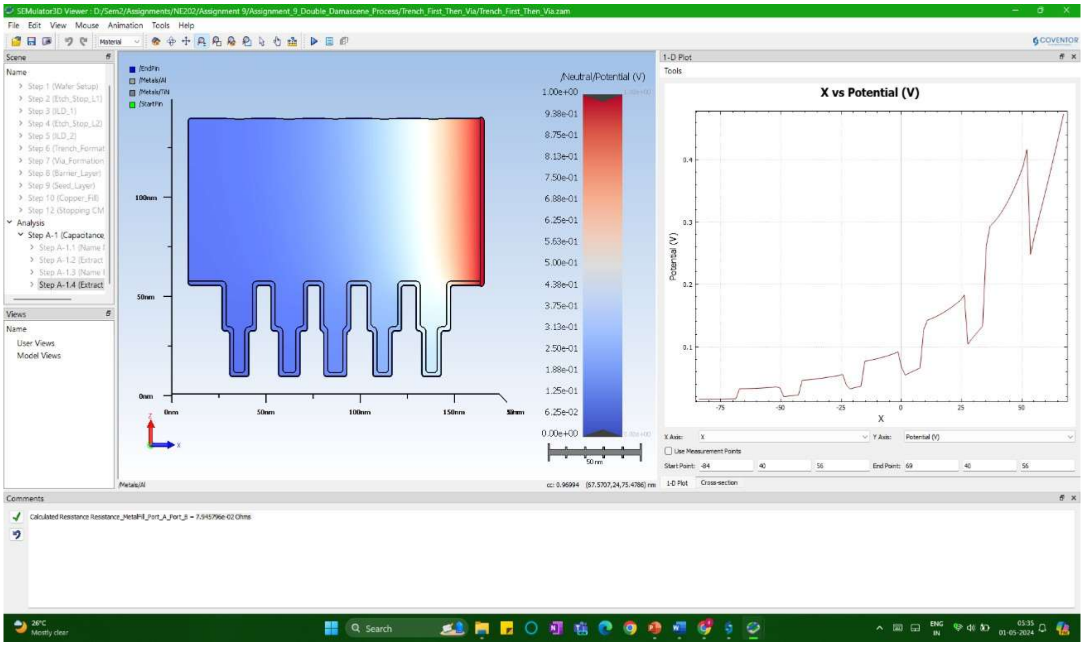


Figure 19: Potential profile for Al

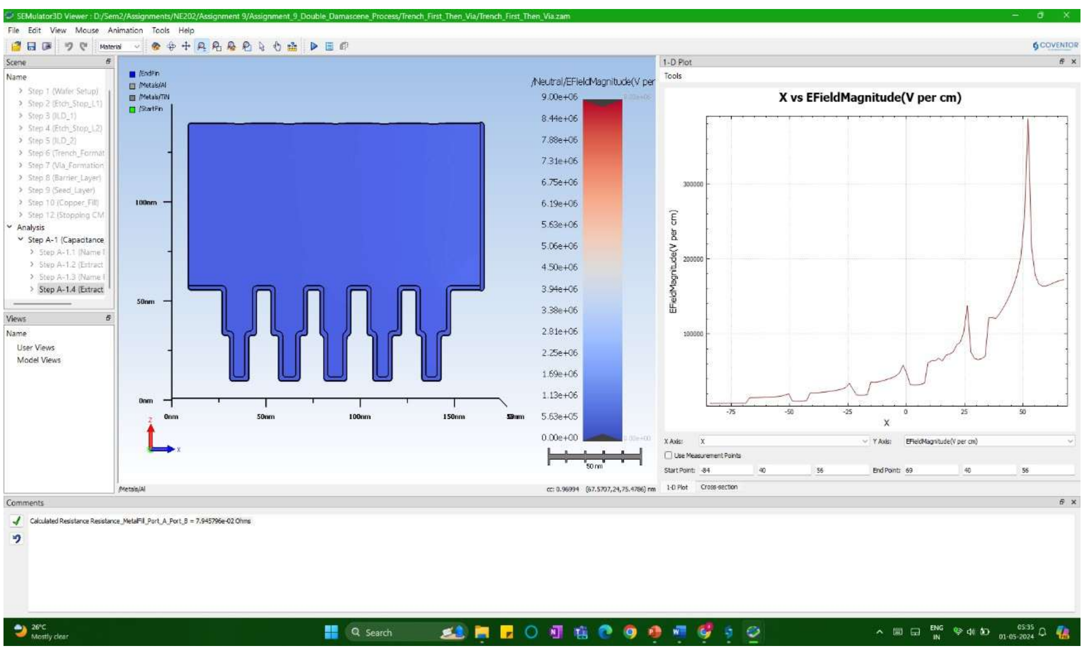


Figure 20: Electric field profile for Al

Changing filler metal to tungsten gives resistance value of 1.482593e-01 Ohms and profiles as shown in figures below,

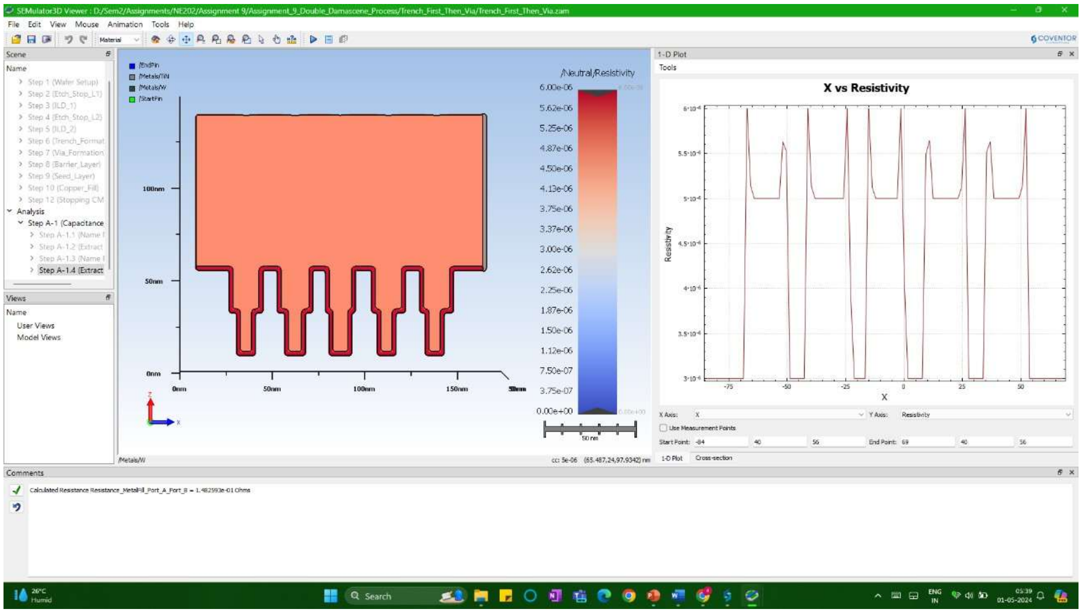


Figure 21 : Resistivity profile for W

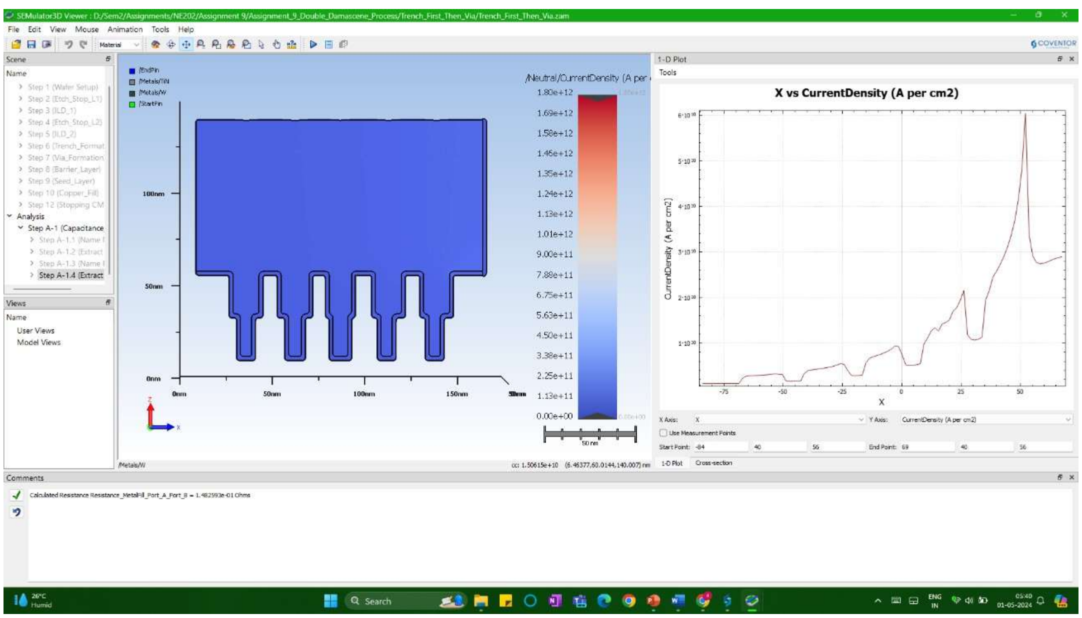


Figure 22 : Current density profile for W

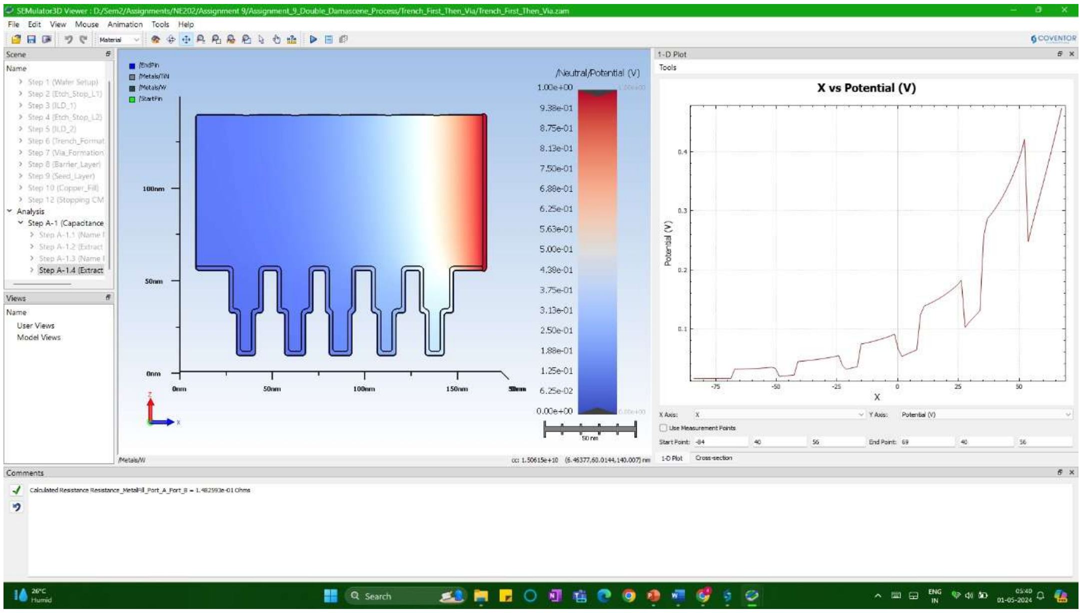


Figure 23 : Potential profile for W

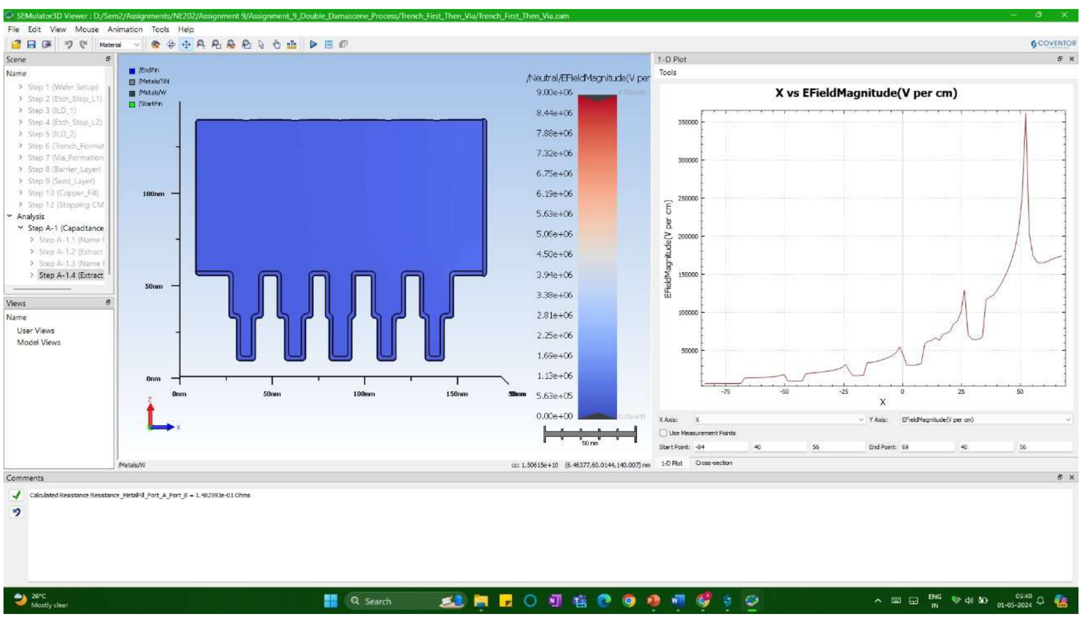


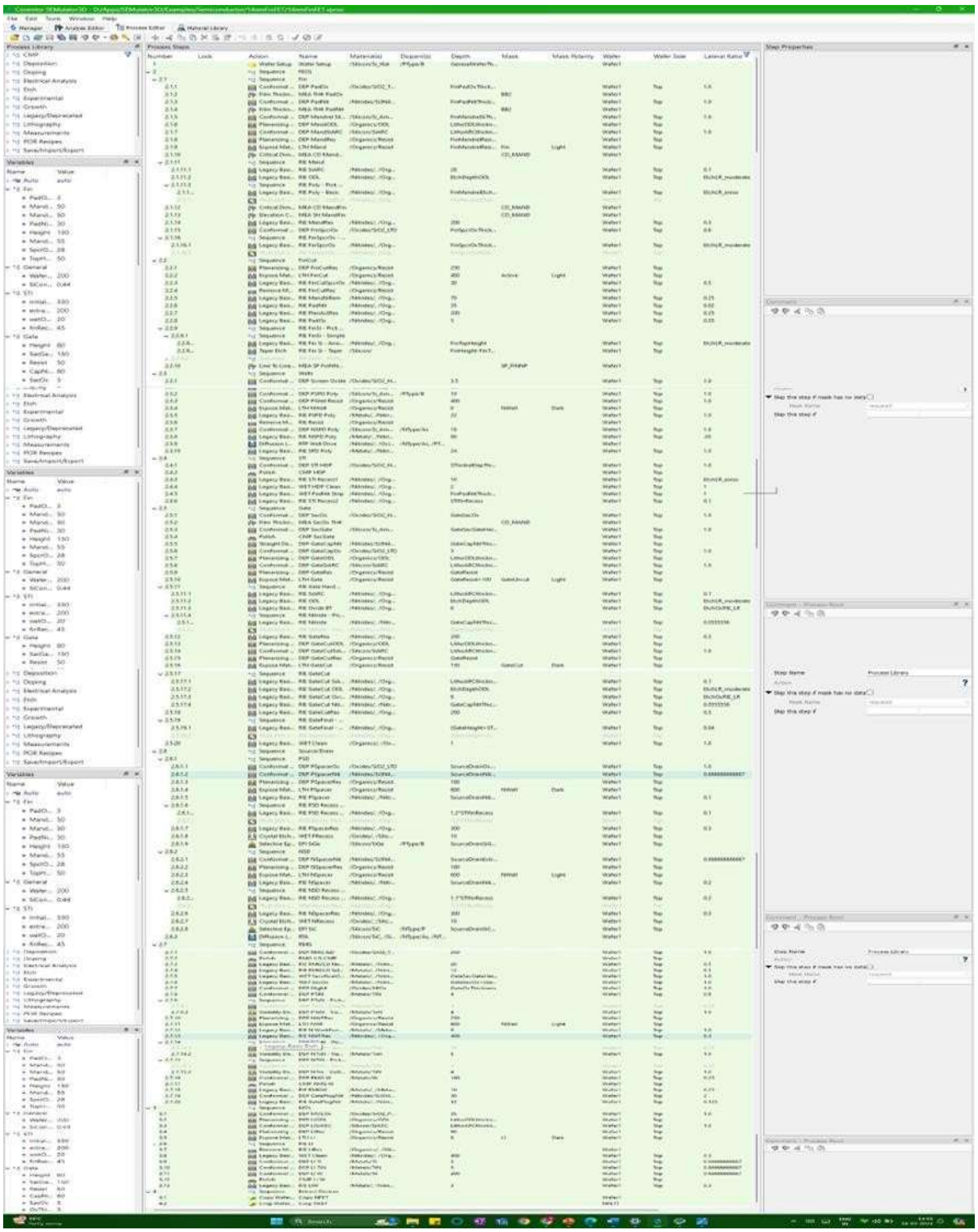
Figure 24 : Electric field profile for W

Copper seems to be best fitting for filler metal applications as it has lowest resistance which will inturn give us fast signal transmission and low loss.

Question E: 14nm advanced logic process

This part deals with 14 nm advanced logic process, we have to identify in which steps metallization is done by double damascene process and which process is used for it and lastly name the material used for different layers.

Figure 25 : Steps involved in FinFET simulation



Resultant sturucture is shown by Figure 26 and Figure 27.

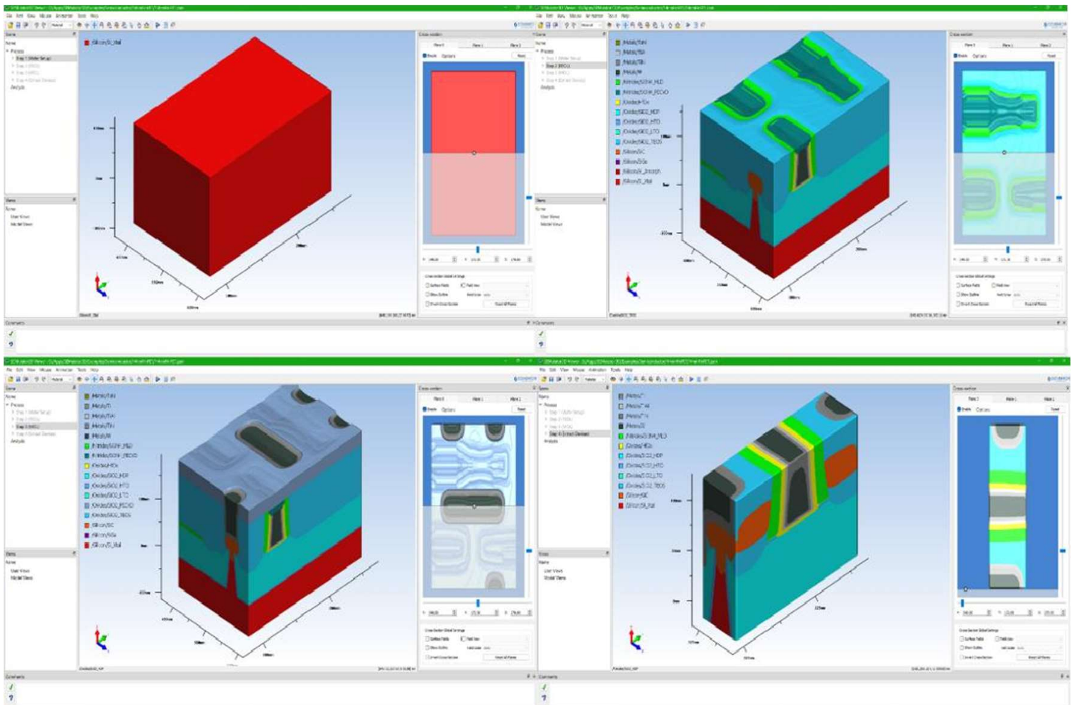


Figure 26 : Simulated 14nm FinFET process (3D view)

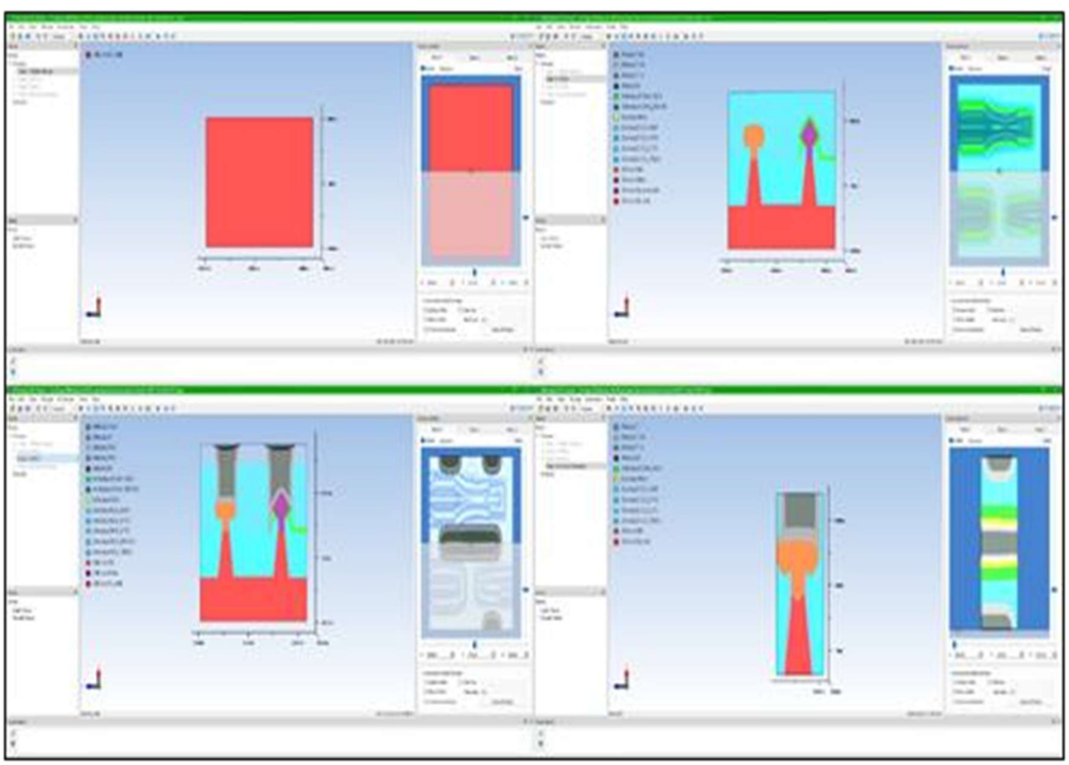


Figure 27 : Simulated 14nm FinFET process (Front View)

Question E(i)

Step 3.6.3 to Step 3.13 is double damascene process first a trench is made followed by IMD, etch stop layer and finally metal deposition (without seed layer).

Question E(ii)

It is trench first process.

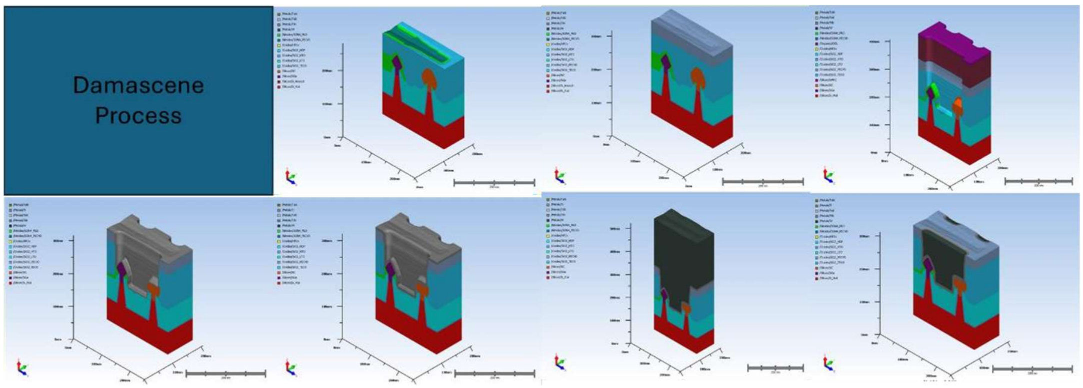


Figure 28 : Trench first Damascene process

Question E(iii)

In this process flow material used for,

- IMD – SiO_2_PECVD.
- etch stop layer – SiO_2_TEOS.
- interconnect metal – Tungsten (W).

Conclusion

We explored via/trench formation, material selection (IMD, etch stop, metal fills), and identified trade-offs. Examining via-first/trench-first approaches, IMD impact on capacitance (favouring low-k), and etch stop function, we determined optimal materials.

References

[1] SEMulator3D/docs/Documentation/Content/Modeling/Electrical/cap_matrix.html